

4. Not everyone is ambiguity averse: Dimmock et al. (2013) show that over one-third of U.S. households are ambiguity seeking, but people who are more risk averse tend to be more uncertainty averse.

5. Picking Up Nickels and Dimes in Front of a Steamroller Redux

Utility functions are a way to measure bad times. Times of low wealth are bad times. Bad times can also be defined relative to an absolute level of safety, a reference point defining gains and losses (across which investors can exhibit different attitudes to risk across gains and losses), past wealth (habit) of an investor, or the relative performance of other investors. During bad times, marginal utility is very high, and the investor considers every spare dollar especially precious.

Mean-variance utility is very restrictive in that bad times are represented only by low portfolio returns and very disperse portfolio returns (low means, high variances). The workhorse utility model of the financial industry is mean-variance. The volatility strategy has a mean and variance very close to the S&P 500, but it has much more pronounced crash risk and fatter tails than the stock market. The volatility strategy's probability distribution function exhibits a very skewed left-hand tail (skewness of -8) and enormous kurtosis (over 100). Mean-variance would not be an appropriate framework to evaluate this strategy, except for those rare (perhaps non-existent) investors who only have preferences over means and variances. Investors exhibiting downside risk aversion would hold much less of this volatility strategy relative to a mean-variance (or approximately CRRA) investor. If deciding between the volatility strategy and a risk-free asset, downside risk averse investors who weight losses about twice as much as gains would optimally choose to forego the volatility strategy and invest only in T-bills.